

CSE 4511: Computer Networks**QUIZ 3****DURATION: 30 MINUTES****WINTER SEMESTER, 2024-2025****FULL MARKS: 20****SET: A****STUDENT ID:**

1. a) Suppose in a custom transport layer protocol named 'AAM', the field to hold the sequence number has 12 bits. What is the minimum and maximum size of the window if it follows the sliding windows technique to send packets?

2
(CO1)
(PO1)

Solution:

Minimum - 1, maximum - 4095 or $2^{12} - 1$

Rubric:

1. For each correct answer, 1 mark

- b) How is it possible to have all 1's in the checksum field? Give proper reasoning

3
(CO1)
(PO1)

Solution:

If the sum of all fields results in all 1s, its complement becomes all 0s. Since a checksum value of all 0s indicates that the checksum is not used, UDP transmits all 1s instead.

Rubric:

1. 1 mark for correct Yes
2. 3 marks for correct reasoning

2. a) What is the significance of Half Close Connection in TCP?

3
(CO1)
(PO2)

Solution:

In Transmission Control Protocol, a half-close connection allows one endpoint to terminate its sending side while still continuing to receive data from the other endpoint. This is achieved using the FIN mechanism, where one side signals that it has finished sending but can still accept incoming data. The significance lies in enabling graceful communication termination, especially in applications like file transfer or streaming, where one side may complete transmission earlier but still needs to receive remaining data without abruptly closing the connection.

- b) Why there are two windows for flow control?

3
(CO1)
(PO2)

Solution:

In Transmission Control Protocol, two windows are used from the perspectives of the sender and receiver to ensure efficient and safe data transmission. The receive window (rwnd) is maintained by the receiver and indicates how much data it can accept based on its available buffer space, thereby preventing buffer overflow. On the other hand, the send window at the sender side represents the amount of data that can be sent without wait-

ing for acknowledgment, which is dynamically adjusted based on the receivers advertised window and network conditions.

The existence of both windows ensures proper coordination: the receiver controls how much it can handle, while the sender regulates how much it injects into the network. This dual-window mechanism enables reliable flow control by matching the senders transmission rate with the receivers capacity.

- c) Why do we need Transmission Sequence Number (TSN) in SCTP even though we have SSN and SI?

3
(CO1)
(PO2)

Solution:

In Stream Control Transmission Protocol, TSN (Transmission Sequence Number) is required to provide reliable data transfer across the entire association, even though SSN (Stream Sequence Number) and SI (Stream Identifier) exist. SSN is used to maintain ordering within individual streams, and SI identifies different streams, but neither ensures overall reliability. TSN enables global sequencing, loss detection, acknowledgment, and retransmission mechanisms across all streams, making it essential for ensuring complete and reliable data delivery in SCTP.

3. A UDP segment has the following fields (all values are in decimal):

6
(CO1)
(PO2)

- Source IP address: 192.168.1.250
- Destination IP address: 192.168.1.251
- Protocol number: 17
- UDP Length: 11 bytes
- Source Port: 65000
- Destination Port: 65010
- Data: 3 bytes with values (200, 100, 50)

Calculate the UDP checksum.

Solution:

Given (decimal):

- Source IP: 192.168.1.250
- Destination IP: 192.168.1.251
- Protocol: 17
- UDP Length: 11 bytes
- Source Port: 65000
- Destination Port: 65010
- Data bytes: (200, 100, 50)

Step 1: Convert to 16-bit binary words

Source IP = 11000000 10101000, 00000001 11111010
Destination IP = 11000000 10101000, 00000001 11111011
Protocol + Zero = 00000000 00010001
UDP Length = 00000000 00001011
Source Port = 11111101 11101000
Destination Port = 11111101 11110010
Data = 11001000 01100100, 00110010 00000000

Step 2: 1's complement addition

Final Sum = 01111011110101110

Step 3: Complement

Checksum = 1000010001010001